

or female professions. Following [Gonen and Goldberg \(2019\)](#), we represent the male and female groups with common names of males and females, rather than with explicitly gendered words (e.g. pronouns). Three tests evaluate the association between a group of male names and a groups of female names to (1) career and family-related words; (2) art and mathematics words; and (3) artistic and scientific fields. In all three tests, we find that the strong association between the groups no longer exists after the projection (non-significant p-values of 0.855, 0.302 and 0.761, respectively).

Bias-by-Neighbors. To measure bias-by-neighbors as discussed in ([Gonen and Goldberg, 2019](#)), we consider the list of professions provided in ([Bolukbasi et al., 2016](#)) and measure the correlation between bias-by projection and bias by neighbors, quantified as the percentage of the top 100 neighbors of each profession which were *originally* biased-by-projection towards either of the genders. We find strong correlation of 0.734 (compared with 0.852 before), indicating that much of the bias-by-neighbors remains.⁵

6.2 Fair Classification: Controlled Setup

We now evaluate using INLP with a deeper classifier, with the goal of achieving fair classification.

Classifier bias measure: TPR-GAP. To measure the bias in a classifier, we follow [De-Arteaga et al. \(2019\)](#) and use the TPR-GAP measure. This measure quantifies the bias in a classifier by considering the difference (GAP) in the True Positive Rate (TPR) between individuals with different protected attributes (e.g. gender/race). The TPR-GAP is tightly related to the notion of fairness by equal opportunity ([Hardt et al., 2016](#)): a fair classifier is expected to show similar success in predicting the task label Y for the two populations, when conditioned on the true class. Formally, for a binary protected attribute z and a true class y , define:

$$TPR_{z,y} = P[\hat{Y} = y | Z = z, Y = y] \quad (2)$$

$$GAP_{z,y}^{TPR} = TPR_{z,y} - TPR_{z',y} \quad (3)$$

where Z is a random variable denoting binary protected attribute, z and z' denote its two values, and

⁵Note that if, for example, STEM-related words are originally biased towards men, the word “chemist” after the projection may still be regarded as male-biased by neighbors, not because an inherent bias but due to its proximity to other *originally* biased words (e.g. other STEM professions).

Ratio	Sentiment		TPR-Gap	
	Original	INLP	Original	INLP
0.5	0.76	0.75	0.19	0.16
0.6	0.78	0.74	0.29	0.22
0.7	0.81	0.66	0.38	0.24
0.8	0.84	0.67	0.45	0.15

Table 1: The Sentiment scores (in accuracy, higher is better) and TPR differences (lower is better) as a function of the ratio of tweets written by black individuals in the positive-sentiment class.

$Y, \hat{Y}$ are random variables denoting the correct class and the predicted class, respectively.

Experiment setup. We begin by experimenting with a controlled setup, where we control for the proportion of the protected attributes within each main-task class. We follow the setup of [Elazar and Goldberg \(2018\)](#) which used a twitter dataset, collected by [Blodgett et al. \(2016\)](#), where each tweet is associated with “race” information and a sentiment which was determined by their belonging to some emoji group.

Naturally, the correlation between the protected class labels and the main-class labels may influence the fairness of the model, as high correlation can encourage the model to condition on the protected attributes. We measure the TPR-GAP on predicting sentiment for the different race groups (African American English (AAE) speakers and Standard American English (SAE) speakers), with different imbalanced conditions, with and without application of our “classifier debiasing” procedure.

In all experiments, the dataset is overly balanced with respect to both sentiment and race (50k instances for each). We change only the *proportion* of each race within each sentiment class (e.g., in the 0.7 condition, the “happy” sentiment class is composed of 70% AAE / 30% SAE, while the “sad” class is composed of 30% AAE / 70% SAE).

Our classifier is based on the DeepMojji encoder ([Felbo et al., 2017](#)), followed by a 1-hidden-layer MLP. The DeepMojji model was trained on millions of tweets in order to predict their emojis; a model which was proven to perform well on different classification tasks ([Felbo et al., 2017](#)), but also encodes demographic information ([Elazar and Goldberg, 2018](#)). We train this classifier to predict sentiment. We then follow the procedure in §5: training a guarding function on the hidden layer of the MLP, and re-training the final linear layer on the guarded vectors. Table 1 presents the results.

		BoW	FastText	BERT
Accuracy (profession)	Original	78.2	78.1	80.9
	+INLP	80.1	73.0	75.2
$GAP_{male}^{TPR,RMS}$	Original	0.203	0.184	0.184
	+INLP	0.124	0.089	0.095

Table 2: Fair classification on the Biographies corpus.

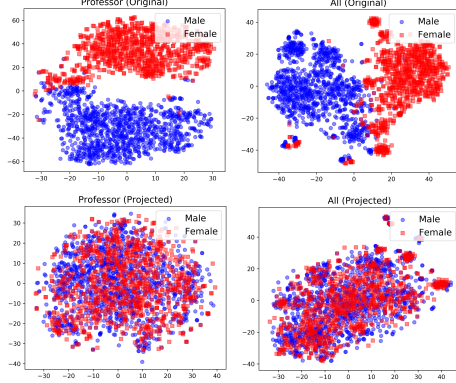


Figure 3: t-SNE projection of BERT representations for the profession “professor” (left) and for a random sample of all professions (right), before and after the projection.

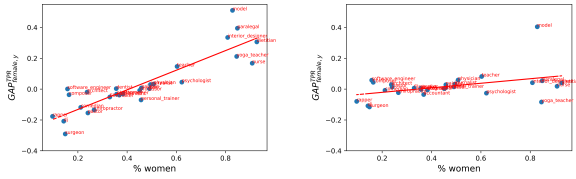


Figure 4: The correlation between $GAP_{female,y}^{TPR}$ and the relative proportion of women in profession y , for BERT representation, before (left; $R=0.883$) and after (right; $R=0.470$) the projection.

As expected the TPR-GAP grows as we increase the correlation between class labels and protected attributes. The accuracy grows as well. Applying our debiasing technique significantly reduced the TPR gap in all settings, although hurting more the main task accuracy in the highly-imbalanced setting. In Appendix A.5, we give some more analysis on the balance between performance and TPR-Gap and show that one can control for this ratio, by using more iterations of INLP.

6.3 Fair Classification: In the Wild

We now evaluate the fair classification approach in a less artificial setting, measuring gender bias in biography classification, following the setup of De-Arteaga et al. (2019).

They scraped the web and collected a dataset of short biographies, annotated by gender and profession. They trained logistic regression classifiers

to predict the profession of the biography’s subject based on three different input representation: bag-of-words (BOW), bag of word-vectors (BWV), and RNN based representation. We repeat their experiments, using INLP for rendering the classifier oblivious of gender.

Setup. Our data contains 393,423 biographies.⁶ We follow the train:dev:test split of De-Arteaga et al. (2019), resulting in 255,710 training examples (65%), 39,369 development examples (10%) and 98,344 (25%) test examples. The dataset has 28 classes (professions), which we predict using a multiclass logistic classifier (in a one-vs-all setting). We consider three input representations: BOW, BWV and BERT (Devlin et al., 2019) based classification. In BOW, we represent each biography as the sum of one-hot vectors, each representing one word in the vocabulary. In the BWV representation, we sum the FastText token representations (Joulin et al., 2017) of the words in the biography. In BERT representation, we represent each biography as the last hidden state of BERT over the *CLS* token. Each of these representations is then fed into the logistic classifier to get final prediction. We do not finetune FastText or BERT.

We run INLP with scikit-learn Pedregosa et al. (2011) linear classifiers. We use 100 logistic classifiers for BOW, 150 linear SVM classifiers for BWV, and 300 linear SVM classifiers for BERT.

Bias measure. We use the TPR-GAP measure for each profession. Following Romanov et al. (2019), we also calculate the root-mean square of $GAP_{g,y}^{TPR}$ over all professions y , to get a single per-gender bias score:

$$GAP_g^{TPR,RMS} = \sqrt{\frac{1}{|C|} \sum_{y \in C} (GAP_{g,y}^{TPR})^2} \quad (4)$$

where C is the set of all labels (professions).

De-Arteaga et al. (2019) have shown that $GAP_{g,y}^{TPR}$ strongly correlates with the percentage of women in profession y , indicating that the true positive rate of the model is influenced by gender.

6.3.1 Results

Main results The results are summarized in Table 2. INLP moderately changes main-task accuracy, with a 1.9% increase in BOW, a 5.1% decrease in performance in BWV and a 5.51% decrease in BERT. $GAP_g^{TPR,RMS}$ is significantly

⁶The original dataset had 399,000 examples, but 5,557 biographies were no longer available on the web.

decreased, indicating that on average, the true positive rate of the classifiers for male and female become closer: in BOW representation, from 0.203 to 0.124 (a 38.91% decrease); in BWV, from 0.184 to 0.089 (a 51.6% decrease); and in BERT, from 0.184 to 0.095 (a 48.36% decrease). We measure the correlation between $GAP_{y,female}^{TPR}$ for each profession y , and the percentage of biographies of women in that profession. In BOW representation, the correlation decreases from 0.894 prior to INLP to 0.670 after it (a 33.4% decrease). In BWV representation, the correlation decreases from 0.896 prior to INLP to 0.425 after it (a 52.5% decrease). In BERT representation, the correlation decreases from 0.883 prior to INLP to 0.470 following it (a 46.7% decreases; Figure 4b). De-Arteaga et al. (2019) report a correlation of 0.71 for BWV representations when using a “scrubbed” version of the biographies, with all pronouns and names removed. INLP significantly outperforms this baseline, while maintaining all explicit gender markers in the input.

Analysis. How does imposing fairness influence the importance the logistic classifier attribute to different words in the biography? We take advantage of the BOW representation and visualize which features (words) influence each prediction (profession), before and after the projection. According to Algorithm 1, to debias an input x , we multiply $W(Px)$. Equivalently, we can first multiply W by P to get a “debaised” weight matrix W' . We begin by testing how much the debaised weights of words that are considered to be biased were changed during the debiasing, compared to random vocabulary words. We compare the relative change before and after the projection of these words, for every occupation. Biased words undergo an average relative change of x1.23 compared to the average change of the entire vocabulary, demonstrating that biased words indeed change more. The per-profession breakout is available in Figure 2 in Appendix §A.6.1.

Next, we test the words that were changed the most during the INLP process. We compare the weight difference before and after the projection. We sort each profession words by weight, and average their location index for each professions. Many words indeed seem gender specific (e.g. *ms.*, *mr.*, *his*, *her*, which appears in locations 1, 2, 3 and 4 respectively), but some seem unrelated, perhaps due to spurious correlations in the data. The complete

list is available in Table 4 in the Appendix §A.6.1; an analogous analysis for the FastText representation is available at Appendix §A.6.2.

7 Limitations

A limitation of our method when used in the context of fairness is that, like other learning approaches, it depends on the data X, Z that is fed to it, and works under the assumption that the training data is sufficiently large and is sampled i.i.d from the same distribution as the test data. This condition is hard to achieve in practice, and failure to provide sufficiently representative training data may lead to biased classifications even after its application. Like other methods, there are no magic guarantees, and the burden of verification remains on the user. It is also important to remember that the method is designed to achieve a very specific sense of protection: removal of linear information regarding a protected attribute. While it may correlate with fairness measures such as demographic parity, it is not designed to ensure them. Finally, it is designed to be fed to a linear decoder, and the attributes are not protected under non-linear classifiers.

8 Conclusion

We present a novel method for removing linearly-represented information from neural representations. We focus on bias and fairness as case studies, and demonstrate that across increasingly complex settings, our method is capable of attenuating societal biases that are expressed in representations learned from data.

While this work focuses on societal bias and fairness, Iterative Nullspace Projection has broader possible use-cases, and can be utilized to remove specific components from a representation, in a controlled and deterministic manner. This method can be applicable for other end goals, such as style-transfer, disentanglement of neural representations and increasing their interpretability. We aim to explore those directions in a future work.

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